SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Concept Mapping (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 2: Concept Mapping
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	S. SRIJITHA	**Reg. No.:**	192471024
Section / Batch:	B. Tech CSE(y Biosciences.	**Date:**	09-10-2026

Code of Conduct

I S-SRIJITHA 192471024 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Srijitha

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Concept Mapping

Q1. An automobile company is developing an autonomous vehicle that uses cameras to detect vehicles, pedestrians, and traffic signs. Objects closer to the camera appear larger, while distant objects appear smaller. **Prepare a concept map applying the image formation and camera projection model to show how a three-dimensional road scene is converted into a two-dimensional image. Include the relationships between focal length, object distance, image distance, and projected image size.**

Q2. A manufacturing company uses cameras to measure the dimensions of bottles moving on a conveyor belt. Engineers observe that bottles at different distances from the camera appear to have different sizes. **Prepare a concept map applying perspective projection to show how focal length, camera position, viewing angle, object distance, and image size are related. Use the concept map to represent how these factors influence the accuracy of bottle measurement.**

Q3. A construction company uses a drone camera to capture images of buildings from different heights and viewing angles. The appearance and measured dimensions of the building change when the drone position is changed.
Prepare a concept map applying image formation and projection concepts to show the relationship between camera orientation, focal length, perspective projection, camera position, calibration, and the accuracy of building measurements.

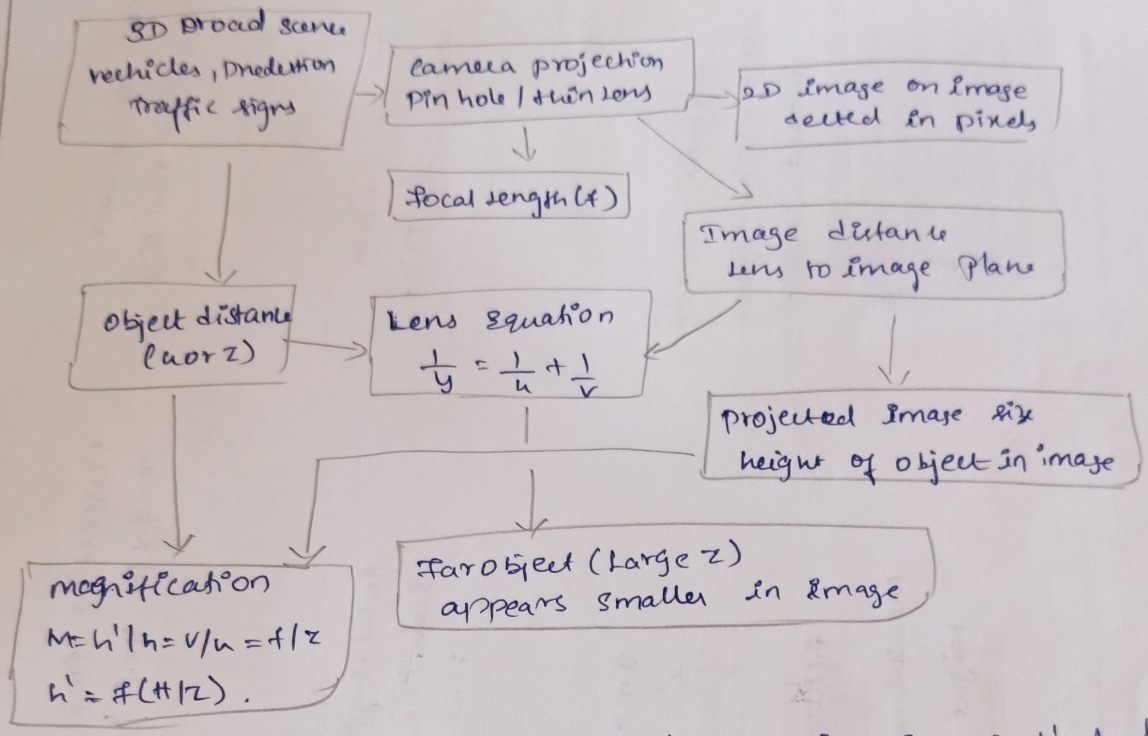
Q4. An industrial company wants to select a suitable projection model for road monitoring and inspection of objects placed on a flat conveyor belt. The engineers are considering perspective and orthographic projection for these applications.
Prepare a concept map comparing perspective and orthographic projection based on depth, object size, distance, parallel lines, and image formation. Use the concept map to identify the appropriate projection model for each application.

Q5. A robotics company wants a robot to identify the position and shape of objects placed on a table using a camera. The camera captures only a two-dimensional representation of the three-dimensional objects.
Prepare a concept map applying the image formation model to show the relationship between a 3D object, camera parameters, projection geometry, focal length, camera orientation, and the resulting 2D image. Show how these concepts help determine the object's position and appearance.

CO1 - AT-2 - Computer vision.

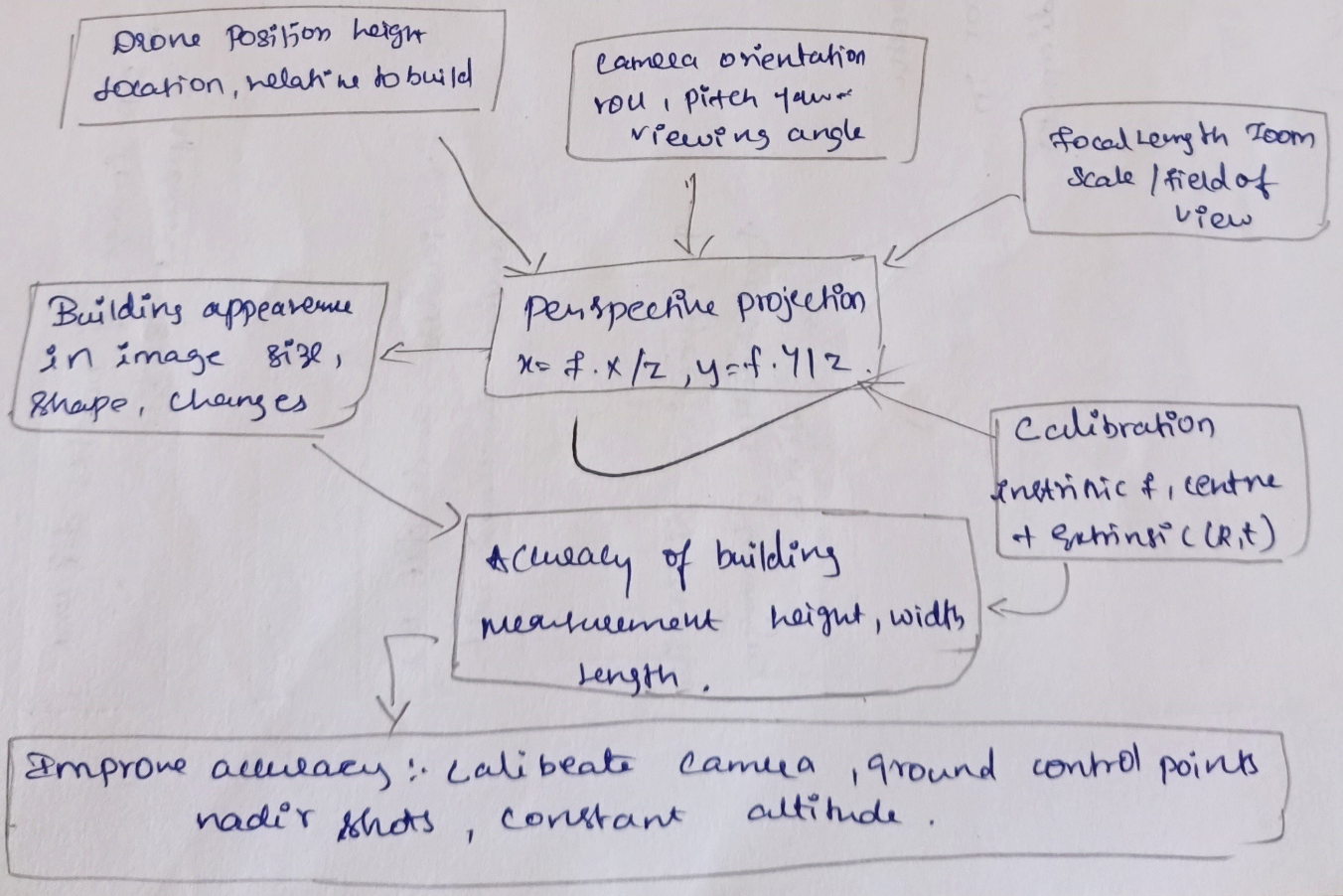
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Question - 1

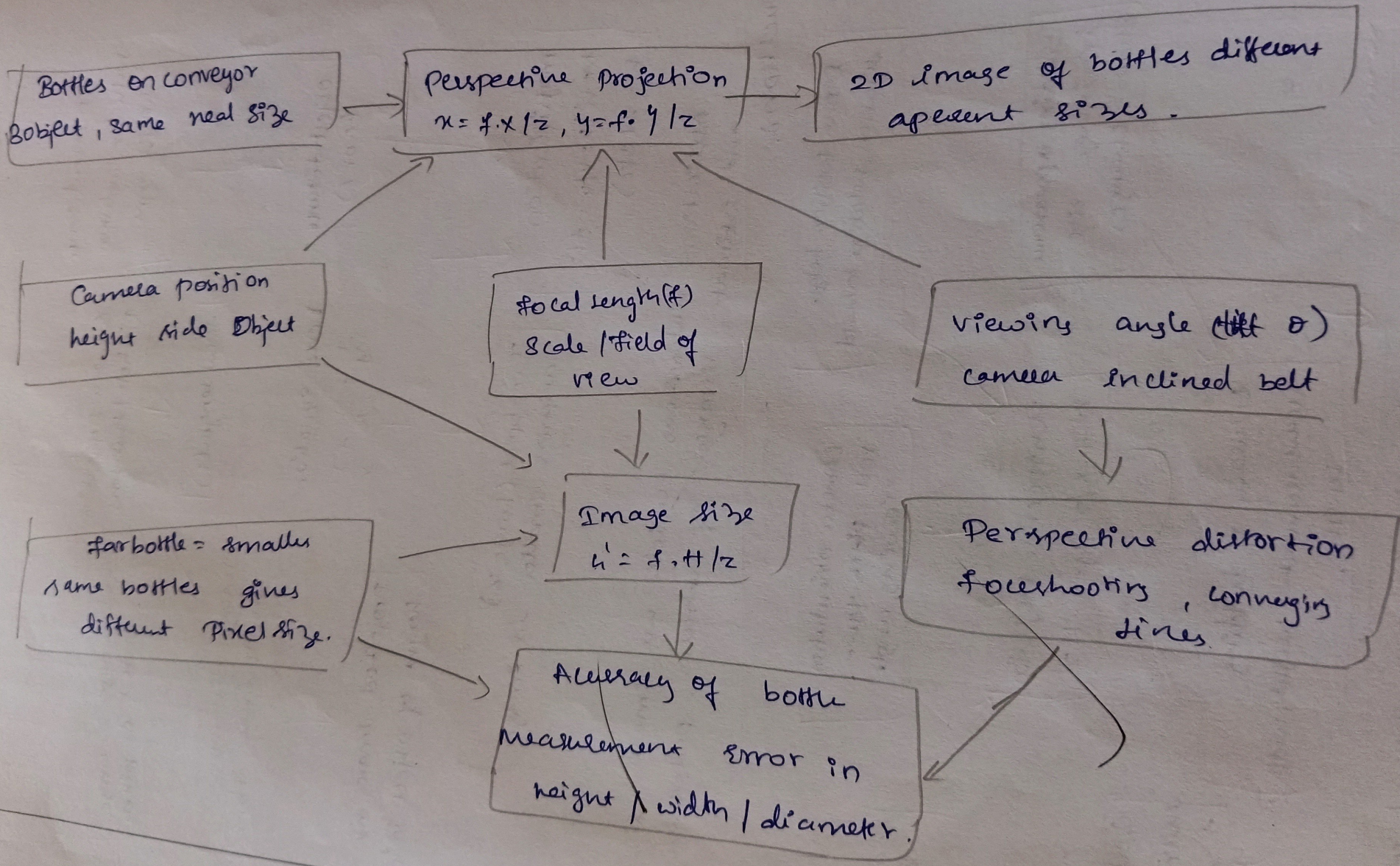


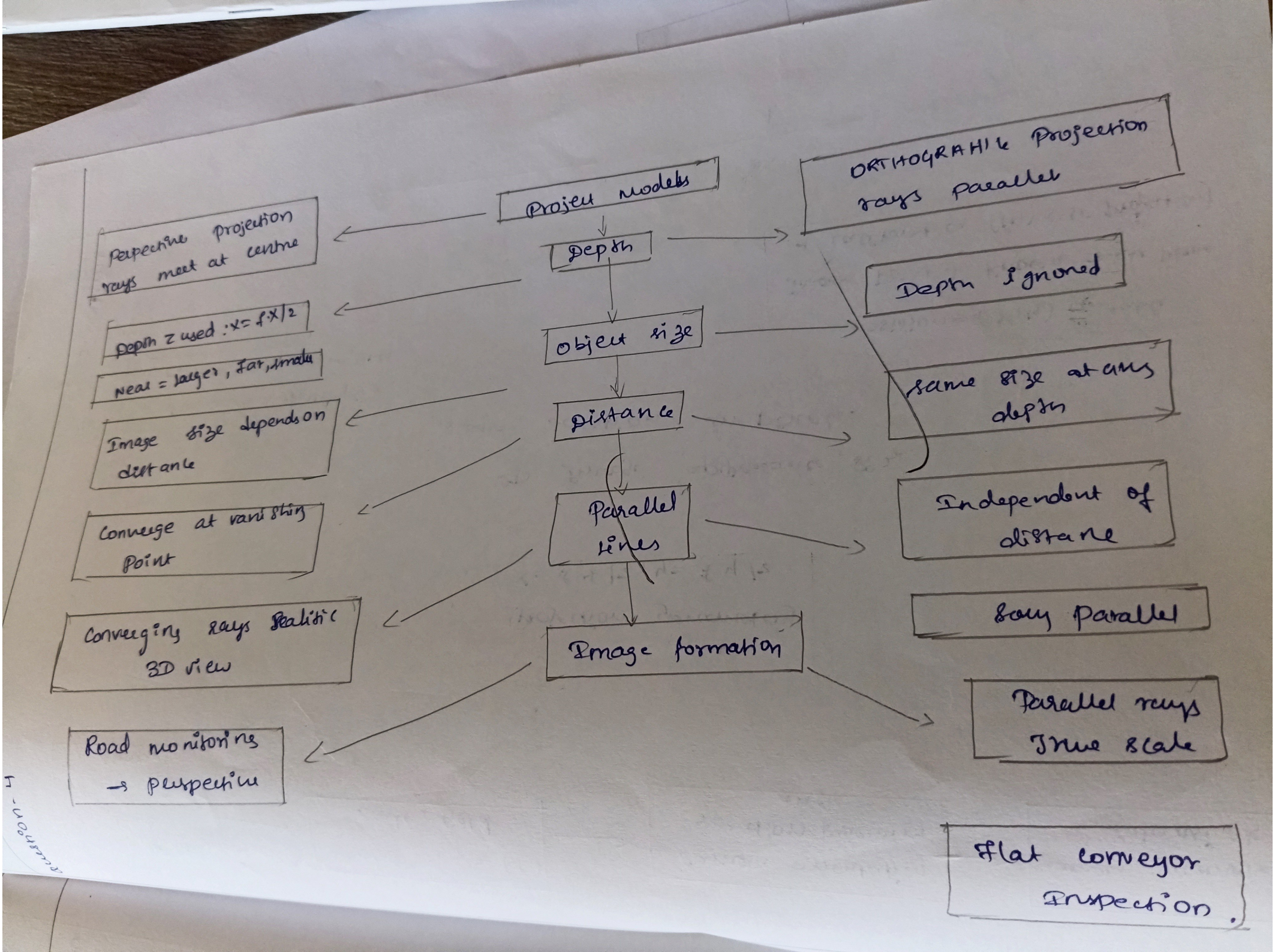
Application:- autonomous vechicles use image size h' to detect and estimate distance of vehicles, pedestrians.

Question 2.

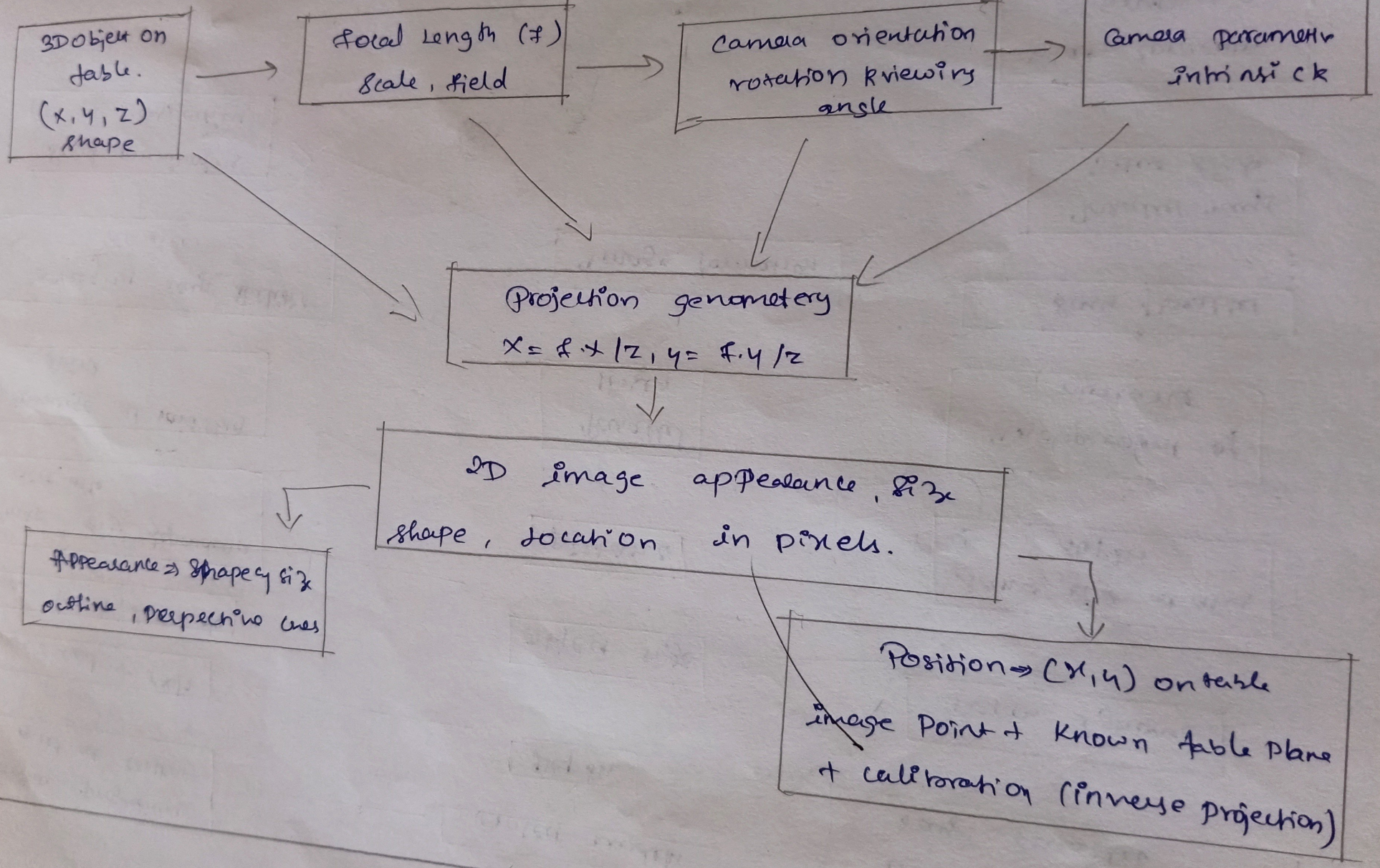


Question-3


Bottles on conveyor
Bobject, same real size
Perspective projection
x = f.X/z, y = f.Y/z
2D image of bottles different apparent sizes.
Camera position
height side Object
focal length(f)
Scale/field of view
viewing angle (θ)
camera inclined belt
Image size
h' = f.H/z
far bottle = smaller
same bottles gives
different Pixel size.
Perspective distortion
foreshooting, converging
lines
Accuracy of bottle
measurement error in
height / width / diameter.


Project Models
Depth
Object size
Distance
Parallel lines
Image formation
Perspective projection rays meet at centre
Depth z used : x = f.X/z
Near = larger, far, smaller
Image size depends on distance
Converge at vanishing point
Converging rays Realistic 3D view
Road monitoring → perspective
ORTHOGRAPHIC Projection rays Parallel
Depth Ignored
Same size at any depth
Independent of distance
Rays Parallel
Parallel rays True scale
Flat conveyor Inspection

Question - 5.


3D Object on table. (x, y, z) shape
Focal length (f) scale, field
Camera orientation rotation viewing angle
Camera parameter intrinsick
Projection geometry x = f.x/z, y = f.y/z
2D image appearance, size shape, location in pixels.
Appearance ⇒ shape, size outline, perspective cues
Position ⇒ (x, y) on table image point + known table plane + calibration (inverse projection)

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: 09/10/25	Date: _____	Date: _____